



# Math Newbie

Learn Math from No Knowledge at All to Confident Beginner

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# Chapter 1

## What Is Math? Meeting Numbers and Counting

### Chapter 1: What Is Math? Meeting Numbers and Counting

Welcome to Math Newbie. If you are just beginning to learn math, you are in the right place. In this chapter, you will discover what math is, why people use it, and how numbers help us understand the world around us.

You will meet the numbers from 0 to 10. You will practice counting objects one at a time. You will count forward, count backward, and learn how a number line can help you see numbers in order.

Do not worry if math feels new. We will go slowly. You can do this step by step.

#### What Is Math?

Math is a way to understand numbers, amounts, shapes, sizes, and patterns.

Math helps us answer questions like:

- How many apples are on the table?
- Do I have enough crayons for everyone?
- What time is it?
- Which tower of blocks is taller?
- How many cookies are left?

Important math word: NUMBER

A number tells how many there are. The number 3 can mean 3 blocks, 3 cats, or 3 pencils.

Important math word: COUNT

To count means to say numbers in order to find out how many objects there are.

Math is not only something you do in school. Math is all around you.

You use math when you:

- Count your toys
- Share snacks with a friend
- Put shoes in pairs
- Look at the clock
- Count steps as you walk

- Find shapes in windows, doors, and books
- Decide who has more or fewer blocks

Math helps you solve little problems every day.

#### Worked Example: Math at Snack Time

Mia has some crackers on a plate. She wants to know how many crackers she has.

She touches each cracker and counts:

1, 2, 3, 4

Mia has 4 crackers.

That is math. She used counting to find the amount.

#### Practice Time 1

Look around you. Find one group of things you can count.

Try counting:

- Fingers on one hand
- Pencils on a desk
- Toys on the floor
- Books on a shelf

Say the numbers slowly. Touch each object as you count it.

#### Meeting Numbers 0 to 10

Numbers can be written as symbols. A symbol is a mark that stands for something.

Here are the number symbols from 0 to 10:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Each number has a name.

0 is zero

1 is one

2 is two



3 is three  
4 is four  
5 is five  
6 is six  
7 is seven  
8 is eight  
9 is nine  
10 is ten

When you see the symbol 5, you can say five. When you see the word five, you can think of the symbol 5.

Important math word: SYMBOL

A symbol is a mark that means something. In math, 7 is the symbol for seven.

Let us connect numbers to objects.

0 means no objects.  
1 means one object.  
2 means two objects.  
3 means three objects.  
4 means four objects.  
5 means five objects.  
6 means six objects.  
7 means seven objects.  
8 means eight objects.  
9 means nine objects.  
10 means ten objects.

Understanding Zero

Zero is a very important number.

Important math word: ZERO

Zero means none or no objects.

If there are no apples in a basket, the basket has 0 apples.

If you eat all 2 cookies on a plate, now there are 0 cookies on the plate.

Zero does not mean something is wrong. It simply means there are none.

Worked Example: Zero Toys

Leo has a toy box. He takes out all the toys and puts them on the rug.

How many toys are left inside the toy box?

There are no toys left inside.

No toys means 0 toys.

So, the toy box has 0 toys inside.

## Practice Time 2

Answer these with a number from 0 to 10.

1. How many noses do you have?
2. How many fingers are on one hand?
3. How many suns do we usually see in the daytime sky?
4. How many elephants are in your room right now?
5. How many toes are on both feet?

Possible answers: 1 nose, 5 fingers, 1 sun, 0 elephants, 10 toes.

## Counting Objects One at a Time

When we count objects, we need to be careful. A good way to count is to touch or point to each object one time.

Important math word: ONE-TO-ONE MATCHING

One-to-one matching means each object gets one number when you count.

Here is how to count carefully:

1. Look at the group of objects.
2. Touch or point to the first object and say 1.
3. Touch or point to the next object and say 2.
4. Keep going in order.
5. Stop when every object has been counted once.
6. The last number you say tells how many objects there are.

This last step is important. If you count 1, 2, 3, 4, the last number is 4. That means there are 4 objects.

### Worked Example: Counting Blocks

There are blocks on a table.

Ava points to each block and counts:

1, 2, 3, 4, 5

The last number Ava says is 5.

So, there are 5 blocks.

Remember:

- Do not skip an object.
- Do not count the same object twice.
- Move slowly.
- Point, touch, or move each object as you count.

### Worked Example: Counting Apples

There are 3 apples in a bowl.

To count them:

1. Point to the first apple. Say 1.
2. Point to the second apple. Say 2.
3. Point to the third apple. Say 3.

The last number is 3.

There are 3 apples.

### Practice Time 3

Use small objects, such as buttons, blocks, beans, or crayons.

Try this:

1. Put 4 objects in a row.
2. Count them by touching each one.
3. Put 6 objects in a row.
4. Count them by touching each one.

5. Put 0 objects in front of you.

6. Say what 0 means.

### Counting Forward

Counting forward means saying numbers in order as they get bigger.

Here are the numbers from 0 to 10:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

When you count forward, you move from a smaller number to a bigger number.

Important math word: FORWARD

Forward means moving ahead in order.

You can count forward when you climb stairs.

Step 1: 1

Step 2: 2

Step 3: 3

Step 4: 4

You can count forward when you put blocks in a tower.

1 block, 2 blocks, 3 blocks, 4 blocks, 5 blocks

Worked Example: Counting Forward from 0

Let us count forward from 0 to 10.

Start at 0.

Then say:

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Now say the whole counting list:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Practice Time 4

Fill in the missing numbers.

1. 0, 1, 2, \_\_, 4
2. 3, 4, \_\_, 6
3. 6, 7, 8, \_\_, 10
4. 1, 2, 3, 4, \_\_

Answers:

1. 3
2. 5
3. 9
4. 5

### Counting Backward

Counting backward means saying numbers in reverse order. The numbers get smaller.

Here are the numbers from 10 back to 0:

10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0

Important math word: BACKWARD

Backward means moving back in order.

Counting backward can help when you think about things going away.

For example, imagine 5 birds sitting on a fence.

If one bird flies away, there are 4 birds.

If another bird flies away, there are 3 birds.

You can count backward:

5, 4, 3

Worked Example: Countdown

A rocket is getting ready to launch.

People count backward:

10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0

Blast off.

Counting backward helps us move down through the numbers.

Worked Example: Toys Going Away

Sam has 4 toy cars on the floor.

He puts one car in the box. Now he has 3 cars on the floor.

He puts another car in the box. Now he has 2 cars on the floor.

The numbers go backward:

4, 3, 2

Practice Time 5

Count backward out loud.

1. Count from 5 to 0.
2. Count from 10 to 0.
3. Fill in the missing numbers: 8, 7, \_\_, 5, \_\_, 3

Answers:

1. 5, 4, 3, 2, 1, 0
2. 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0
3. 6 and 4

Using a Simple Number Line

A number line is a picture that shows numbers in order.

Important math word: NUMBER LINE

A number line is a line with numbers placed in order.

Here is a simple number line from 0 to 10:

0 -- 1 -- 2 -- 3 -- 4 -- 5 -- 6 -- 7 -- 8 -- 9 -- 10

A number line helps you see where numbers are.

When you count forward, you move to the right:

0 to 1 to 2 to 3 to 4 to 5

When you count backward, you move to the left:

5 to 4 to 3 to 2 to 1 to 0

Worked Example: Moving Forward on a Number Line

Start at 2.

Move forward one space. You land on 3.

Move forward one more space. You land on 4.

So, counting forward from 2 gives:

2, 3, 4

Worked Example: Moving Backward on a Number Line

Start at 6.

Move backward one space. You land on 5.

Move backward one more space. You land on 4.

So, counting backward from 6 gives:

6, 5, 4

Practice Time 6

Use this number line:

0 -- 1 -- 2 -- 3 -- 4 -- 5 -- 6 -- 7 -- 8 -- 9 -- 10

Answer the questions.

1. What number comes after 4?
2. What number comes after 7?
3. What number comes before 5?

4. What number comes before 1?
5. If you start at 0 and count forward three spaces, where do you land?

Answers:

1. 5
2. 8
3. 4
4. 0
5. 3

### Everyday Counting Practice

You can practice counting almost anywhere. Try these simple activities with a grown-up, a friend, or by yourself.

#### Activity 1: Count Your Steps

1. Stand in one place.
2. Take one step and say 1.
3. Take another step and say 2.
4. Keep counting until you reach 10.

Then try counting backward from 10 to 0 as you take small steps back.

#### Activity 2: Snack Count

Use small snacks, such as crackers or grapes.

1. Put 5 snacks on a plate.
2. Count them one at a time.
3. Eat one snack.
4. Count how many are left.

You are using math to keep track of an amount.

#### Activity 3: Number Hunt

Look for numbers around you.

You might see numbers on:

- A clock



- A calendar
- A book page
- A door
- A phone
- A sign

When you find a number from 0 to 10, say its name.

Mini Challenge: Can You Find It?

Find these numbers somewhere near you:

- 1
- 2
- 5
- 10

Point to each number and say its name.

Beginner Word Problems

A word problem is a math question told with words. Do not worry. We will keep them very simple.

Important math word: WORD PROBLEM

A word problem is a story that asks a math question.

To solve a counting word problem:

1. Listen to the story.
2. Find what is being counted.
3. Count carefully.
4. Say the answer.

Worked Example 1

Lena has 3 toy bears. How many toy bears does Lena have?

Step 1: What are we counting? Toy bears.

Step 2: What number is in the story? 3.

Step 3: Lena has 3 toy bears.

Answer: 3 toy bears.

## Worked Example 2

There are no bananas in the bowl. How many bananas are in the bowl?

Step 1: What are we counting? Bananas.

Step 2: Are there any bananas? No.

Step 3: No bananas means 0 bananas.

Answer: 0 bananas.

## Practice Time 7

Try these word problems.

1. Ben has 2 red blocks. How many red blocks does Ben have?
2. There are 5 birds in a tree. How many birds are in the tree?
3. Kim has no pencils on her desk. How many pencils are on her desk?
4. A box has 10 crayons. How many crayons are in the box?

Answers:

1. 2 red blocks
2. 5 birds
3. 0 pencils
4. 10 crayons

## Friendly Recap

Great work. You have started your math journey.

In this chapter, you learned that:

- Math helps us understand numbers, amounts, shapes, sizes, and patterns.
- We use math every day when we count, share, check time, and solve problems.
- Numbers can be written as symbols.
- The numbers from 0 to 10 are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10.
- Zero means none or no objects.
- Counting means saying numbers in order.
- One-to-one matching means each object gets one number.
- Counting forward means the numbers get bigger.
- Counting backward means the numbers get smaller.
- A number line shows numbers in order and helps you count.

Before you move on, try this final practice.

### Final Chapter 1 Practice

1. Count forward from 0 to 10.
2. Count backward from 10 to 0.
3. Count 7 small objects.
4. Show 0 objects and explain what zero means.
5. Use a number line to find the number after 6.
6. Use a number line to find the number before 3.

Answers:

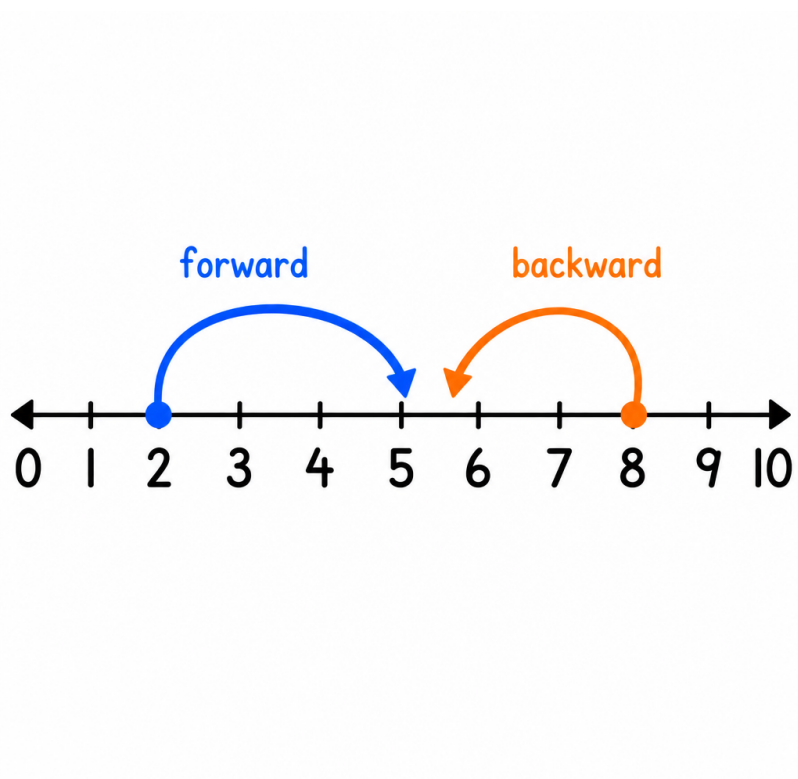
1. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
2. 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0
3. You should count 7 objects one at a time.
4. Zero means none or no objects.
5. The number after 6 is 7.
6. The number before 3 is 2.

You are now ready to keep building your math skills. Next, you will learn how to compare numbers and understand which amounts are more, fewer, or the same.



*Two children explore everyday math by counting blocks, looking at shapes, sharing snacks, and noticing a clock.*

*A number line from 0 to 10 with large numbers and a friendly character ready to count forward.*



*A number line from 0 to 10 with one arrow counting forward and another arrow counting backward.*

# Chapter 2

## Bigger, Smaller, and Place Value

### Chapter 2: Bigger, Smaller, and Place Value

In Chapter 1, you learned that numbers help us count things. In this chapter, you will discover how to compare numbers. That means you will learn how to tell which number is bigger, which number is smaller, or when two numbers are the same.

You will also learn a very important math idea called place value. Place value helps us understand numbers like 12, 24, and 35. These numbers are made from tens and ones.

Do not worry if that sounds new. We will go slowly, use simple examples, and practice together.

### Math Words to Know

Here are some important words for this chapter:

- More means a larger amount.
- Fewer means a smaller amount.
- Same means the amounts match.
- Equal means the amounts are the same.
- Bigger number means a number that is worth more.
- Smaller number means a number that is worth less.
- Greater than means bigger than.
- Less than means smaller than.
- Tens are groups of 10.
- Ones are single pieces left over after making tens.

### Comparing Groups of Objects

Let's begin with things we can see and count.

Imagine there are two plates of cookies.

Plate A has 3 cookies.

Plate B has 5 cookies.

Which plate has more cookies?

Let's count:

Plate A: 1, 2, 3

Plate B: 1, 2, 3, 4, 5

Plate B has more cookies because 5 is more than 3.

Plate A has fewer cookies because 3 is fewer than 5.

Worked Example 1: More or Fewer?

There are 4 red blocks and 6 blue blocks.

Step 1: Count the red blocks.

Red blocks: 1, 2, 3, 4

Step 2: Count the blue blocks.

Blue blocks: 1, 2, 3, 4, 5, 6

Step 3: Compare the numbers.

6 is bigger than 4.

Answer: There are more blue blocks. There are fewer red blocks.

Same Amounts and Equal Groups

Sometimes two groups have the same amount.

Example:

A bowl has 4 grapes.

Another bowl has 4 grapes.

Both bowls have 4 grapes. The amounts are the same.

We can also say the amounts are equal.

Equal means the same amount.

Worked Example 2: Are They Equal?

There are 7 stars on one paper and 7 stars on another paper.

Step 1: Count the first group.

First paper: 7 stars

Step 2: Count the second group.

Second paper: 7 stars

Step 3: Compare.

7 and 7 are the same.

Answer: The two groups are equal.

Practice Time 1

Count and compare each pair.

1. There are 2 cats and 5 dogs. Which group has more?
2. There are 6 pencils and 6 crayons. Are the groups equal?
3. There are 9 buttons and 4 beads. Which group has fewer?
4. There are 3 bananas and 8 apples. Which group has more?

Answers:

1. Dogs have more.
2. Yes, they are equal.
3. Beads have fewer.
4. Apples have more.

Bigger and Smaller Numbers

We can compare numbers even when we do not see objects.

Look at these numbers:

2 and 7

Which number is bigger?

7 is bigger than 2.

Which number is smaller?

2 is smaller than 7.

A helpful idea is this: when we count forward, numbers usually get bigger.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

The farther along the counting line a number is, the bigger it is.

For example:

- 9 is bigger than 6.
- 4 is smaller than 8.
- 10 is bigger than 1.
- 5 is equal to 5.

Using a Number Line

A number line is a line with numbers in order.

Here is a simple number line:

0 - 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 10

Numbers to the right are bigger.

Numbers to the left are smaller.

Look at 3 and 8 on the number line.

0 - 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9 - 10

8 is to the right of 3, so 8 is bigger than 3.

3 is to the left of 8, so 3 is smaller than 8.

Worked Example 3: Compare with a Number Line

Which is bigger, 5 or 9?

Step 1: Find 5 on the number line.

Step 2: Find 9 on the number line.

Step 3: See which number is farther to the right.

9 is farther to the right than 5.

Answer: 9 is bigger than 5.

Practice Time 2



Use the number line in your mind, or draw one from 0 to 10.

1. Which is bigger: 4 or 6?
2. Which is smaller: 8 or 2?
3. Which is bigger: 10 or 7?
4. Are 5 and 5 equal?

Answers:

1. 6 is bigger.
2. 2 is smaller.
3. 10 is bigger.
4. Yes, 5 and 5 are equal.

Greater Than, Less Than, and Equal To

Now let's learn three math comparison symbols.

- $>$  means greater than.
- $<$  means less than.
- $=$  means equal to.

Greater than means bigger than.

Less than means smaller than.

Equal to means the same as.

Here are examples:

$8 > 3$  means 8 is greater than 3.

$2 < 6$  means 2 is less than 6.

$5 = 5$  means 5 is equal to 5.

A Friendly Trick for the Symbols

The greater than and less than symbols can look like a little open mouth.

The open mouth likes to face the bigger number.

Example:

$9 > 4$

The open side faces 9 because 9 is bigger.

Example:

$$3 < 7$$

The open side faces 7 because 7 is bigger.

Remember:

- The open side faces the bigger number.
- The pointy side points to the smaller number.

Worked Example 4: Choose the Right Symbol

Compare 6 and 2.

Step 1: Ask, which number is bigger?  
6 is bigger than 2.

Step 2: Choose the symbol with the open side facing 6.

Answer:  $6 > 2$

Worked Example 5: Equal Numbers

Compare 4 and 4.

Step 1: Look at both numbers.  
They are the same.

Step 2: Use the equal sign.

Answer:  $4 = 4$

Practice Time 3

Fill in the missing symbol:  $>$ ,  $<$ , or  $=$ .

1.  $7 \underline{\hspace{1cm}} 3$

2.  $1 \underline{\hspace{1cm}} 5$

3.  $9 \underline{\hspace{1cm}} 9$

4.  $4 \underline{\hspace{1cm}} 8$

5.  $10 \underline{\hspace{0.5cm}} 6$

Answers:

1.  $7 > 3$

2.  $1 < 5$

3.  $9 = 9$

4.  $4 < 8$

5.  $10 > 6$

Place Value: Tens and Ones

Now we are ready for a big beginner math idea: place value.

Place value means that a digit has a value because of where it is placed in a number.

A digit is one number symbol, like 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9.

In the number 7, there is just one digit.

But in the number 27, there are two digits: 2 and 7.

The 2 and the 7 do not mean the same thing. Their places tell us what they are worth.

The number 27 means:

- 2 tens

- 7 ones

A ten is a group of 10.

A one is a single item.

So 27 means 2 groups of ten and 7 extra ones.

Let's build it:

$$10 + 10 = 20$$

20 and 7 more ones makes 27.

That is why 27 is made of 2 tens and 7 ones.

Building Numbers with Tens and Ones

Let's use blocks in our imagination.

A long stick of 10 blocks is one ten.

A single block is one one.

To build 14:

- Use 1 ten.
- Use 4 ones.

So  $14 = 1 \text{ ten and } 4 \text{ ones}$ .

To build 23:

- Use 2 tens.
- Use 3 ones.

So  $23 = 2 \text{ tens and } 3 \text{ ones}$ .

To build 35:

- Use 3 tens.
- Use 5 ones.

So  $35 = 3 \text{ tens and } 5 \text{ ones}$ .

Worked Example 6: What Are the Tens and Ones?

Look at the number 42.

Step 1: Look at the first digit.

The first digit is 4. It is in the tens place.

So there are 4 tens.

Step 2: Look at the second digit.

The second digit is 2. It is in the ones place.

So there are 2 ones.

Answer: 42 has 4 tens and 2 ones.

Worked Example 7: Build the Number

Build the number 16.

Step 1: Look at the tens digit.

The 1 tells us there is 1 ten.

Step 2: Look at the ones digit.

The 6 tells us there are 6 ones.

Step 3: Put them together.

1 ten and 6 ones make 16.

Answer:  $16 = 1$  ten and 6 ones.

#### Practice Time 4

Write the tens and ones.

1.  $12 =$  \_\_\_ ten and \_\_\_ ones

2.  $24 =$  \_\_\_ tens and \_\_\_ ones

3.  $35 =$  \_\_\_ tens and \_\_\_ ones

4.  $41 =$  \_\_\_ tens and \_\_\_ one

5.  $50 =$  \_\_\_ tens and \_\_\_ ones

Answers:

1.  $12 = 1$  ten and 2 ones

2.  $24 = 2$  tens and 4 ones

3.  $35 = 3$  tens and 5 ones

4.  $41 = 4$  tens and 1 one

5.  $50 = 5$  tens and 0 ones

#### Comparing Two-Digit Numbers

Place value can help us compare bigger numbers too.

Look at 24 and 35.

24 has 2 tens and 4 ones.

35 has 3 tens and 5 ones.

Which number is bigger?

35 is bigger because it has more tens. Three tens is more than two tens.

So:

$$35 > 24$$

$$24 < 35$$

Here is another example:

Compare 18 and 12.

18 has 1 ten and 8 ones.

12 has 1 ten and 2 ones.

They have the same number of tens. So we compare the ones.

8 ones is more than 2 ones.

So 18 is bigger than 12.

$$18 > 12$$

#### Worked Example 8: Compare Two-Digit Numbers

Compare 31 and 29.

Step 1: Look at the tens.

31 has 3 tens.

29 has 2 tens.

Step 2: Compare the tens.

3 tens is more than 2 tens.

Step 3: Choose the bigger number.

31 is bigger.

Answer:  $31 > 29$

#### Worked Example 9: Same Tens, Different Ones

Compare 46 and 43.

Step 1: Look at the tens.

Both numbers have 4 tens.

Step 2: Look at the ones.

46 has 6 ones.

43 has 3 ones.

Step 3: Compare the ones.

6 ones is more than 3 ones.

Answer:  $46 > 43$

### Practice Time 5

Compare the numbers. Fill in  $>$ ,  $<$ , or  $=$ .

1.  $12 \underline{\hspace{1cm}} 15$

2.  $28 \underline{\hspace{1cm}} 21$

3.  $33 \underline{\hspace{1cm}} 33$

4.  $40 \underline{\hspace{1cm}} 19$

5.  $25 \underline{\hspace{1cm}} 52$

Answers:

1.  $12 < 15$

2.  $28 > 21$

3.  $33 = 33$

4.  $40 > 19$

5.  $25 < 52$

### Try It with Objects

You can practice place value with simple things at home or in class.

You can use:

- Sticks
- Straws
- Blocks
- Buttons
- Beans
- Small toys
- Drawn circles on paper

Make groups of 10. Each group of 10 is one ten. Then count the extra ones.

### Activity: Build 23

Step 1: Count out 10 small objects.

Step 2: Put them in one group. That is 1 ten.

Step 3: Count out another 10 small objects.

Step 4: Put them in another group. Now you have 2 tens.

Step 5: Count out 3 extra objects. These are 3 ones.

You built 23.

$23 = 2 \text{ tens and } 3 \text{ ones.}$

### Activity: Build and Compare

Build these numbers using objects or drawings:

1. 14

2. 18

3. 22

4. 31

Then answer:

- Which number is the smallest?

- Which number is the biggest?

Answer:

14 is the smallest.

31 is the biggest.

### Quick Challenge

Use what you know about comparing and place value.

1. Which has more: 2 tens and 5 ones, or 3 tens and 1 one?

2. What number is 4 tens and 2 ones?

3. What number is 1 ten and 9 ones?

4. Fill in the symbol:  $36 \underline{\hspace{1cm}} 34$

5. Fill in the symbol:  $20 \underline{\hspace{1cm}} 20$

Answers:



1. 3 tens and 1 one is more. It is 31, and 31 is greater than 25.
2. 42
3. 19
4.  $36 > 34$
5.  $20 = 20$

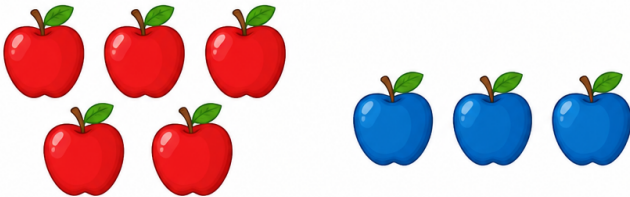
### Friendly Recap

Great work! In this chapter, you learned how to compare numbers and begin using place value.

Remember these big ideas:

- More means a larger amount.
- Fewer means a smaller amount.
- Equal means the same amount.
- A bigger number is worth more.
- A smaller number is worth less.
- On a number line, numbers to the right are bigger.
- The symbol  $>$  means greater than.
- The symbol  $<$  means less than.
- The symbol  $=$  means equal to.
- Two-digit numbers are made of tens and ones.
- In 24, the 2 means 2 tens and the 4 means 4 ones.

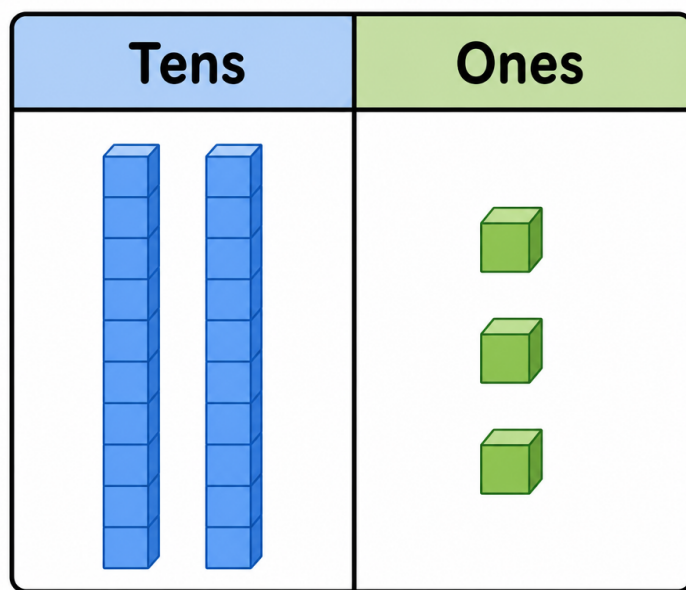
You are learning to look carefully at numbers and understand what they mean. That is a big step in math. In the next chapter, you will use numbers to put amounts together with addition and take amounts away with subtraction.

**1**  
  
**more**      **fewer**

**2**  
  
**equal**

**3**  


Three panels compare groups of objects to show more, fewer, equal, and that 7 is greater than 4.



23

The number 23 shown as two ten-rods in the tens column and three single cubes in the ones column.

# Chapter 3

## Adding and Subtracting: Putting Together and Taking Away

### CHAPTER 3

#### Adding and Subtracting: Putting Together and Taking Away

In this chapter, you will discover two very important math actions: addition and subtraction.

Addition helps us put things together and find out how many we have in all. Subtraction helps us take things away and find out how many are left.

You already know how to count. Now you will use counting to solve small math problems. We will use toys, fruit, pencils, fingers, pictures, and number lines. Go slowly. Count carefully. You can do this!

#### 1. Addition Means Putting Together

Imagine you have 3 toy cars. Your friend gives you 2 more toy cars.

Now you have more cars than before. To find out how many cars you have in all, you can count them together:

1, 2, 3, 4, 5

So, 3 cars and 2 cars make 5 cars.

**MATH WORD:** Addition means putting groups together or getting more.

When we add, the number usually gets bigger.

Example:

You have 4 apples.

You get 1 more apple.

Now you have 5 apples.

That is addition.

Picture idea:

3 toy cars + 2 toy cars = 5 toy cars

## 2. The Plus Sign

The plus sign looks like this:

+

MATH WORD: Plus means add.

When you see the plus sign, it tells you to put numbers together.

Read this number sentence:

$$3 + 2 = 5$$

You can read it like this:

3 plus 2 equals 5.

It means:

3 things and 2 more things make 5 things.

## 3. The Equals Sign

The equals sign looks like this:

=

MATH WORD: Equals means is the same as.

The equals sign tells us that both sides match.

$$3 + 2 = 5$$

The left side says 3 plus 2.

The right side says 5.

They match because 3 and 2 together make 5.

Key reminder:

- + means put together.

- = means is the same as.

- The answer in an addition problem is called the sum.

MATH WORD: Sum means the answer to an addition problem.

## 4. Simple Addition with Objects

Using real objects is a great way to learn addition. You can use blocks, buttons, crayons, spoons, shells, or your fingers.

#### Worked Example 1: Pencils

Mia has 2 pencils.

She finds 3 more pencils.

How many pencils does Mia have now?

Steps:

1. Count the first group: 1, 2.
2. Count the second group: 1, 2, 3.
3. Put the groups together.
4. Count all the pencils: 1, 2, 3, 4, 5.

Number sentence:

$$2 + 3 = 5$$

Mia has 5 pencils.

#### Worked Example 2: Bananas

There are 4 bananas on the table.

Dad puts 2 more bananas on the table.

How many bananas are there now?

Steps:

1. Start with 4 bananas.
2. Add 2 more bananas.
3. Count all of them: 1, 2, 3, 4, 5, 6.

Number sentence:

$$4 + 2 = 6$$

There are 6 bananas.

#### Practice Time: Add with Objects

Use your fingers or small objects to solve.

1.  $1 + 2 = \underline{\quad}$

2.  $3 + 1 = \underline{\quad}$

3.  $2 + 2 = \underline{\quad}$

4.  $4 + 1 = \underline{\quad}$

5.  $5 + 2 = \underline{\quad}$

Check by counting all the objects together.

## 5. Using Pictures for Addition

Pictures can help you see addition. You can draw circles, stars, apples, or simple lines.

Example:

Draw 3 circles.

Draw 2 more circles.

Count all the circles.

○○○○○

Count: 1, 2, 3, 4, 5

Number sentence:

$$3 + 2 = 5$$

Worked Example 3: Stars

There are 5 stars in one group.

There are 1 star in another group.

How many stars are there in all?

Draw:

\*\*\*\*\*

Count all the stars:

1, 2, 3, 4, 5, 6

Number sentence:

$$5 + 1 = 6$$

There are 6 stars in all.

Practice Time: Draw and Add

Draw pictures to help you.

1. Draw 2 apples. Draw 3 more apples. How many apples in all? \_\_\_\_

2. Draw 4 balls. Draw 2 more balls. How many balls in all? \_\_\_\_

3. Draw 1 flower. Draw 5 more flowers. How many flowers in all? \_\_\_\_

## 6. Using Fingers for Addition

Your fingers are helpful math tools.

Example:

What is  $3 + 4$ ?

Steps:

1. Hold up 3 fingers on one hand.
2. Hold up 4 fingers on the other hand.
3. Count all the fingers that are up.

1, 2, 3, 4, 5, 6, 7

So:

$$3 + 4 = 7$$

Finger tip:

If you are adding small numbers, fingers can help you count carefully.

Practice Time: Add with Fingers

1.  $2 + 4 = \underline{\quad}$
2.  $5 + 1 = \underline{\quad}$
3.  $3 + 3 = \underline{\quad}$
4.  $6 + 2 = \underline{\quad}$

## 7. Using a Number Line for Addition

A number line is a line with numbers in order.

0 1 2 3 4 5 6 7 8 9 10

MATH WORD: Number line means a line that shows numbers in order.

For addition, we jump forward on the number line.

Worked Example 4:  $4 + 3$

Steps:

1. Start at 4.
2. Jump forward 3 times.
3. Say the numbers as you jump: 5, 6, 7.
4. Land on 7.

So:

$$4 + 3 = 7$$

Worked Example 5:  $2 + 5$

Steps:

1. Start at 2.
2. Jump forward 5 times.
3. Count the jumps: 3, 4, 5, 6, 7.
4. Land on 7.

So:

$$2 + 5 = 7$$

Remember:

For addition, jump to the right. The numbers get bigger.

Practice Time: Number Line Addition

Use a number line from 0 to 10.

1. Start at 3. Jump forward 2. Where do you land? \_\_\_\_
2. Start at 5. Jump forward 3. Where do you land? \_\_\_\_
3. Start at 1. Jump forward 4. Where do you land? \_\_\_\_
4. Start at 6. Jump forward 1. Where do you land? \_\_\_\_

## 8. Subtraction Means Taking Away

Now let us learn subtraction.

Imagine there are 6 cookies on a plate. You eat 2 cookies. How many cookies are left?

Start with 6 cookies.

Take away 2 cookies.

Count what is left:

1, 2, 3, 4

So, 6 take away 2 is 4.

MATH WORD: Subtraction means taking away or finding what is left.

When we subtract, the number usually gets smaller.

Picture idea:



6 cookies - 2 cookies = 4 cookies

## 9. The Minus Sign

The minus sign looks like this:

-

MATH WORD: Minus means subtract or take away.

Read this number sentence:

$$6 - 2 = 4$$

You can read it like this:

6 minus 2 equals 4.

It means:

Start with 6. Take away 2. You have 4 left.

Key reminder:

- - means take away.

- = means is the same as.

- The answer in a subtraction problem is called the difference.

MATH WORD: Difference means the answer to a subtraction problem.

## 10. Simple Subtraction with Objects

Objects make subtraction easy to see. Put some objects in front of you. Then move some away.

Worked Example 6: Crayons

Sam has 5 crayons.

He gives 2 crayons to his sister.

How many crayons does Sam have left?

Steps:

1. Start with 5 crayons.

2. Take away 2 crayons.

3. Count the crayons left: 1, 2, 3.

Number sentence:

$$5 - 2 = 3$$

Sam has 3 crayons left.

### Worked Example 7: Toy Blocks

There are 7 blocks on the floor.

You put 3 blocks in a box.

How many blocks are still on the floor?

Steps:

1. Start with 7 blocks.
2. Take away 3 blocks.
3. Count what is left: 1, 2, 3, 4.

Number sentence:

$$7 - 3 = 4$$

There are 4 blocks still on the floor.

### Practice Time: Subtract with Objects

Use small objects or your fingers.

1.  $4 - 1 = \underline{\quad}$

2.  $5 - 3 = \underline{\quad}$

3.  $6 - 2 = \underline{\quad}$

4.  $7 - 4 = \underline{\quad}$

5.  $8 - 1 = \underline{\quad}$

Check by starting with the first number, taking away the second number, and counting what is left.

### 11. Using Pictures for Subtraction

You can draw pictures for subtraction too. Draw the starting number. Cross out the number being taken away. Count what is not crossed out.

### Worked Example 8: Fish

There are 5 fish.

2 fish swim away.

How many fish are left?

Draw 5 fish:

><> ><> ><> ><> ><>

Cross out 2 fish. Count the fish not crossed out.

3 fish are left.

Number sentence:

$$5 - 2 = 3$$

Worked Example 9: Balloons

Lena has 6 balloons.

1 balloon pops.

How many balloons are left?

Steps:

1. Draw 6 balloons.
2. Cross out 1 balloon.
3. Count the balloons left: 1, 2, 3, 4, 5.

Number sentence:

$$6 - 1 = 5$$

Lena has 5 balloons left.

Practice Time: Draw and Subtract

Draw pictures. Cross out the ones that go away.

1. Draw 5 cookies. Cross out 2. How many are left? \_\_\_\_
2. Draw 7 stars. Cross out 3. How many are left? \_\_\_\_
3. Draw 4 flowers. Cross out 1. How many are left? \_\_\_\_

## 12. Using a Number Line for Subtraction

For subtraction, we jump backward on the number line.

0 1 2 3 4 5 6 7 8 9 10

Worked Example 10:  $8 - 3$

Steps:

1. Start at 8.
2. Jump backward 3 times.
3. Say the numbers as you jump: 7, 6, 5.
4. Land on 5.

So:

$$8 - 3 = 5$$

## Worked Example 11: $6 - 4$

Steps:

1. Start at 6.
2. Jump backward 4 times.
3. Count the jumps: 5, 4, 3, 2.
4. Land on 2.

So:

$$6 - 4 = 2$$

Remember:

For subtraction, jump to the left. The numbers get smaller.

### Practice Time: Number Line Subtraction

Use a number line from 0 to 10.

1. Start at 5. Jump backward 2. Where do you land? \_\_\_\_
2. Start at 9. Jump backward 3. Where do you land? \_\_\_\_
3. Start at 7. Jump backward 1. Where do you land? \_\_\_\_
4. Start at 10. Jump backward 5. Where do you land? \_\_\_\_

## 13. Addition and Subtraction Stories

Many math problems are little stories. Listen for what is happening.

If things are joining together or getting more, use addition.

If things are being taken away or leaving, use subtraction.

Helpful words for addition:

- more
- in all
- altogether
- join
- get

Helpful words for subtraction:

- left
- take away
- gave away
- lost
- went away

### Worked Story Problem 1: Addition

Nora has 3 stickers.

Her teacher gives her 4 more stickers.

How many stickers does Nora have in all?

Think:

Nora gets more stickers, so we add.

Steps:

1. Start with 3.
2. Add 4 more.
3. Count: 4, 5, 6, 7.

Number sentence:

$$3 + 4 = 7$$

Nora has 7 stickers.

### Worked Story Problem 2: Subtraction

Ben has 8 grapes.

He eats 3 grapes.

How many grapes are left?

Think:

Ben eats some grapes, so we subtract.

Steps:

1. Start with 8.
2. Take away 3.
3. Count what is left or jump back: 7, 6, 5.

Number sentence:

$$8 - 3 = 5$$

Ben has 5 grapes left.

### 14. Mini Word Problems

Try these. Read slowly. Ask yourself: Are things being put together, or are things being taken away?

1. Ava has 2 red blocks and 3 blue blocks. How many blocks does Ava have in all? \_\_\_\_
2. There are 6 birds in a tree. 2 birds fly away. How many birds are left? \_\_\_\_

3. You have 4 shells. You find 4 more shells. How many shells do you have now? \_\_\_\_
4. There are 9 carrots. A rabbit eats 5 carrots. How many carrots are left? \_\_\_\_
5. Leo has 1 pencil. His friend gives him 6 more pencils. How many pencils does Leo have? \_\_\_\_

## 15. Checking Your Answer

Good math learners check their work. Checking means looking again to make sure your answer makes sense.

Ways to check:

- Count the objects again.
- Count the picture again.
- Use your fingers.
- Use a number line.
- Ask: Did the number get bigger or smaller?

For addition, your answer usually gets bigger.

Example:

$$2 + 5 = 7$$

7 is bigger than 2 and bigger than 5.

For subtraction, your answer usually gets smaller.

Example:

$$9 - 4 = 5$$

5 is smaller than 9.

Be careful:

If you are subtracting, do not add by mistake.

If you are adding, do not take away by mistake.

## 16. Mixed Practice

Now let us mix addition and subtraction. Solve each one. You may use objects, pictures, fingers, or a number line.

1.  $2 + 1 = \underline{\quad}$

2.  $5 - 2 = \underline{\quad}$

3.  $4 + 3 = \underline{\quad}$

4.  $6 - 1 = \underline{\quad}$

5.  $3 + 5 = \underline{\quad}$

6.  $8 - 4 = \underline{\quad}$

7.  $1 + 7 = \underline{\quad}$

8.  $10 - 2 = \underline{\quad}$

### Challenge Practice:

1. There are 4 cups on a table. Mom puts 2 more cups on the table. How many cups are there now? \_\_\_\_
2. There are 7 kids on the playground. 3 kids go inside. How many kids are still on the playground? \_\_\_\_
3. You have 5 toy animals. You get 3 more. How many toy animals do you have in all? \_\_\_\_
4. There are 6 cupcakes. 2 cupcakes are eaten. How many cupcakes are left? \_\_\_\_

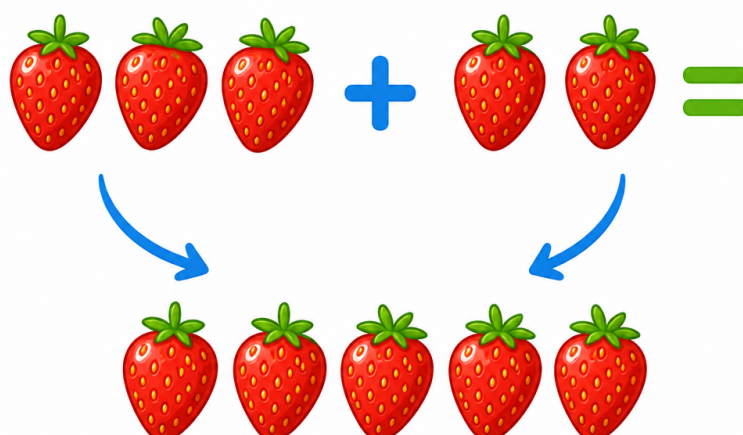
### Friendly Recap

Great work! In this chapter, you learned how to add and subtract.

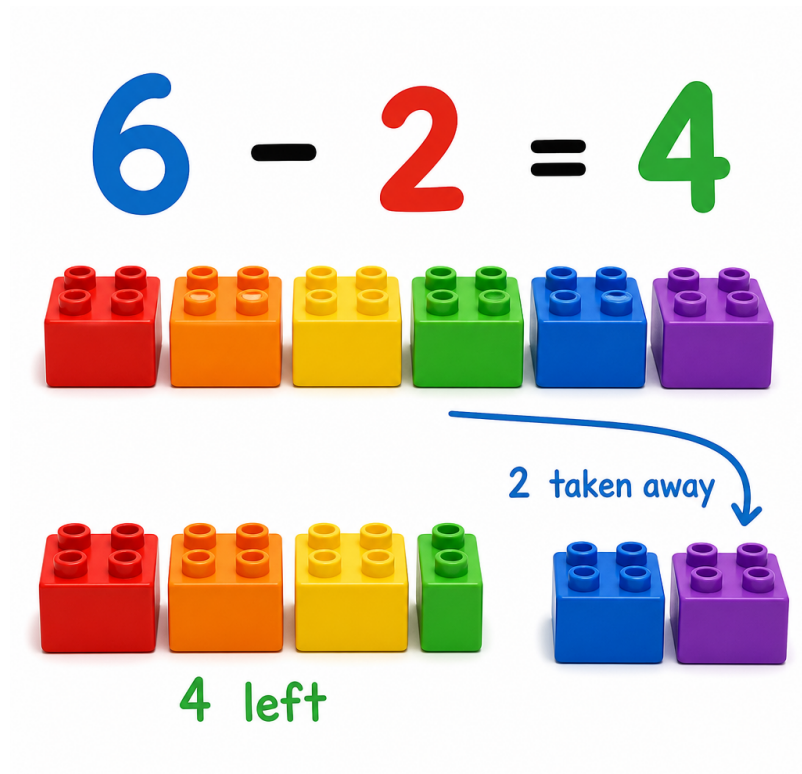
Remember these big ideas:

- Addition means putting together or getting more.
- The plus sign + tells us to add.
- The equals sign = means is the same as.
- The answer to an addition problem is called the sum.
- Subtraction means taking away or finding what is left.
- The minus sign - tells us to subtract.
- The answer to a subtraction problem is called the difference.
- For addition on a number line, jump forward.
- For subtraction on a number line, jump backward.
- You can use objects, pictures, fingers, and number lines to help you.
- Always check your answer by counting carefully.

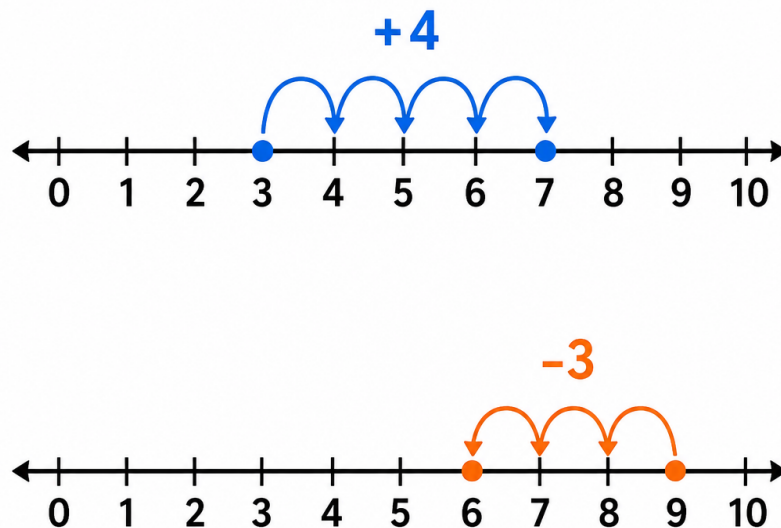
You are now learning how numbers can change. Sometimes numbers grow when we add. Sometimes numbers shrink when we subtract. In the next chapter, you will keep building your math power with new ways to think about groups.



Three strawberries and two strawberries are put together to make five strawberries.



Six toy blocks are shown, with two being taken away and four blocks left.



Two number lines show four forward jumps from 3 to 7 and three backward jumps from 9 to 6.



# Chapter 4

## Groups, Sharing, Shapes, Patterns, and Measurement

### Chapter 4: Groups, Sharing, Shapes, Patterns, and Measurement

In this chapter, you will discover new ways to use math. You already know that numbers can help us count, add, and subtract. Now we will use numbers to make equal groups, share things fairly, name shapes, continue patterns, and measure things around us.

Do not worry if these ideas feel new. We will take small steps. We will use pictures in our minds, simple examples, and practice questions.

#### 1. Multiplication Means Equal Groups

Important Math Word: Multiplication means counting equal groups.

An equal group is a group that has the same number of things as another group.

Look at these groups of stars:

Group 1: star star

Group 2: star star

Group 3: star star

Each group has 2 stars.

There are 3 equal groups.

We can count all the stars like this:

$$2 + 2 + 2 = 6$$

So, 3 groups of 2 stars make 6 stars.

This is multiplication.

Multiplication is a faster way to add the same number again and again.

Important Math Word: Repeated addition means adding the same number more than one time.

### Example 1: Bags of Apples

Mia has 4 bags.

Each bag has 3 apples.

We can think:

Bag 1 has 3 apples.

Bag 2 has 3 apples.

Bag 3 has 3 apples.

Bag 4 has 3 apples.

Add them:

$$3 + 3 + 3 + 3 = 12$$

So, 4 groups of 3 apples make 12 apples.

Sometimes people write this as  $4 \times 3 = 12$ .

The  $\times$  sign is called the multiplication sign. For now, just remember that it means groups of.

$4 \times 3$  means 4 groups of 3.

### How to Solve a Multiplication Problem with Groups

Let us solve: 3 groups of 4.

Step 1: Draw 3 groups.

Group 1: circle circle circle circle

Group 2: circle circle circle circle

Group 3: circle circle circle circle

Step 2: Count how many are in each group.

Each group has 4.

Step 3: Add the groups.

$$4 + 4 + 4 = 12$$

Step 4: Say the answer.

3 groups of 4 is 12.

Practice Time: Equal Groups

1. There are 2 groups of 5 dots. How many dots are there in all?
2. There are 3 groups of 3 cookies. How many cookies are there in all?
3. There are 4 groups of 2 blocks. How many blocks are there in all?

Try drawing the groups before you answer.

Check Your Answers

1.  $5 + 5 = 10$
2.  $3 + 3 + 3 = 9$
3.  $2 + 2 + 2 + 2 = 8$

2. Counting by Groups

When groups are equal, we can count by the group number.

If each group has 2, we can count:

2, 4, 6, 8, 10

This is called counting by 2s.

If each group has 5, we can count:

5, 10, 15, 20

This is called counting by 5s.

Counting by groups helps us count faster.

Example 2: Pairs of Socks

A pair means 2.

If you have 5 pairs of socks, you can count by 2s:

2, 4, 6, 8, 10

So, 5 pairs of socks have 10 socks in all.

### Practice Time: Count by Groups

Fill in the missing numbers.

1. Count by 2s: 2, 4, \_\_, 8, \_\_
2. Count by 5s: 5, \_\_, 15, \_\_
3. Count by 10s: 10, 20, \_\_, 40

### Check Your Answers

1. 2, 4, 6, 8, 10
2. 5, 10, 15, 20
3. 10, 20, 30, 40

### 3. Division Means Sharing Equally

Important Math Word: Division means sharing into equal parts or equal groups.

When we divide, we make fair shares.

Important Math Word: Fair share means each person or group gets the same amount.

Let us imagine 8 cupcakes and 4 friends.

We want to share the cupcakes fairly.

Friend 1 gets 1 cupcake.

Friend 2 gets 1 cupcake.

Friend 3 gets 1 cupcake.

Friend 4 gets 1 cupcake.

We used 4 cupcakes. There are 4 cupcakes left.

Now give one more to each friend.

Friend 1 gets 2 cupcakes in all.

Friend 2 gets 2 cupcakes in all.

Friend 3 gets 2 cupcakes in all.

Friend 4 gets 2 cupcakes in all.

Now all 8 cupcakes are shared.

Each friend gets 2 cupcakes.

So, 8 shared by 4 makes 2 each.

Sometimes people write this as  $8 \div 4 = 2$ .

The  $\div$  sign is called the division sign. For now, remember that it means shared by.

### How to Solve a Sharing Problem

Let us solve: Share 6 stickers equally between 3 children.

Step 1: Draw 3 children or 3 circles.

Child 1: empty

Child 2: empty

Child 3: empty

Step 2: Give 1 sticker to each child.

Child 1: sticker

Child 2: sticker

Child 3: sticker

You used 3 stickers. You have 3 left.

Step 3: Give 1 more sticker to each child.

Child 1: sticker sticker

Child 2: sticker sticker

Child 3: sticker sticker

Step 4: Count each share.

Each child has 2 stickers.

So, 6 shared by 3 makes 2 each.

### Practice Time: Sharing Equally

Draw circles to show the people or groups. Then share the objects one at a time.

1. Share 4 bananas between 2 monkeys. How many does each monkey get?
2. Share 9 buttons between 3 boxes. How many buttons go in each box?
3. Share 10 crackers between 5 children. How many crackers does each child get?

Check Your Answers

1. Each monkey gets 2 bananas.
2. Each box gets 3 buttons.
3. Each child gets 2 crackers.

Remember: Sharing is fair only when each group gets the same number.

#### 4. Multiplication and Division Are Helpers

Multiplication and division are connected.

Multiplication makes equal groups and counts the total.

Division starts with the total and shares it into equal groups.

Example:

3 groups of 2 apples make 6 apples.

Now turn it around:

6 apples shared into 3 equal groups makes 2 apples in each group.

Both ideas use equal groups.

#### Key Ideas to Remember

- Multiplication: equal groups make a total.
- Division: a total is shared into equal groups.
- Drawing pictures can help you solve both.
- Fair shares must be equal.

#### 5. Basic Shapes

Math is not only about numbers. Math also helps us name and describe shapes.

Important Math Word: Shape means the form or outline of something.

You can find shapes everywhere: in toys, signs, books, windows, plates, doors, and clocks.

## Circle

A circle is round.

It has no corners.

It has no straight sides.

Examples of circles:

- A ball
- A clock face
- A plate
- A wheel

## Square

A square has 4 equal sides.

It has 4 corners.

Examples of squares:

- A square window
- A game board space
- A sticky note
- Some floor tiles

## Triangle

A triangle has 3 sides.

It has 3 corners.

Examples of triangles:

- A slice of pizza
- A triangle sign
- A roof shape
- A musical triangle

## Rectangle

A rectangle has 4 sides.

It has 4 corners.

Two sides are longer, and two sides are shorter.

Examples of rectangles:

- A door
- A book
- A phone
- A table top

Shape Sorting

Important Math Word: Sort means to put things into groups that are alike.

You can sort shapes by asking questions:

- Is it round?
- Does it have corners?
- How many sides does it have?
- Are the sides all the same length?

Example 3: Sort the Objects

Look at these objects:

ball, door, pizza slice, clock, book, square tile

We can sort them like this:

Circles: ball, clock

Rectangles: door, book

Triangle: pizza slice

Square: square tile

Practice Time: Shapes

Name the shape.

1. I am round and have no corners. What am I?
2. I have 3 sides and 3 corners. What am I?
3. I have 4 equal sides. What am I?
4. I look like a door and have 4 corners. What am I?



## Check Your Answers

1. Circle
2. Triangle
3. Square
4. Rectangle

## Shape Hunt Activity

Look around your room. Find one object for each shape.

- Circle: \_\_\_\_\_
- Square: \_\_\_\_\_
- Triangle: \_\_\_\_\_
- Rectangle: \_\_\_\_\_

You can do this with a friend or grown-up. Take turns naming shapes.

## 6. Patterns

Important Math Word: Pattern means something that repeats in a way we can notice.

Patterns can use colors, shapes, numbers, sounds, or movements.

Patterns help us guess what comes next.

### Color Pattern

Red, blue, red, blue, red, blue

What comes next?

Red comes next.

The pattern is red, blue, red, blue.

### Shape Pattern

Circle, square, circle, square, circle, square

What comes next?

Circle comes next.

## Number Pattern

2, 4, 6, 8

What comes next?

10 comes next because we are counting by 2s.

## Example 4: Find the Repeating Part

Look at this pattern:

Triangle, triangle, circle, triangle, triangle, circle

The repeating part is:

Triangle, triangle, circle

So the next three shapes are:

Triangle, triangle, circle

## How to Continue a Pattern

Step 1: Look carefully.

Step 2: Find what repeats.

Step 3: Say the repeating part out loud.

Step 4: Add what comes next.

## Practice Time: Patterns

What comes next?

1. Star, heart, star, heart, star, \_\_
2. Circle, circle, square, circle, circle, square, \_\_
3. 5, 10, 15, 20, \_\_
4. Clap, stomp, clap, stomp, clap, \_\_

## Check Your Answers

1. Heart
2. Circle

3. 25

4. Stomp

Make Your Own Pattern

Choose colors, shapes, or numbers.

Here are some ideas:

- Red, yellow, red, yellow
- Square, square, circle, square, square, circle
- 1, 2, 1, 2

Now make your own:

\_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

## 7. Simple Measurement

Important Math Word: Measurement means finding out the size, length, amount, or time of something.

Measurement helps us answer questions like:

- How long is it?
- How tall is it?
- How heavy is it?
- What time is it?
- Which object is bigger or smaller?

We use measurement every day.

Length with Rulers

Important Math Word: Length means how long something is from one end to the other.

A ruler helps us measure length.

You can use a ruler to measure:

- A pencil
- A crayon
- A book

- A toy car

When you measure with a ruler, start at the beginning mark. Then look at where the object ends.

### Beginner Measuring Steps

Step 1: Put the start of the ruler at one end of the object.

Step 2: Keep the ruler straight.

Step 3: Look at the number near the other end of the object.

Step 4: Say the length.

### Example 5: Measuring a Pencil

A pencil starts at 0 on the ruler.

The pencil ends at 7.

The pencil is about 7 units long.

A unit is what we count with when measuring. On many rulers, the units are inches or centimeters. For now, just practice matching the object to the numbers.

### Measuring with Everyday Objects

You can measure even without a ruler.

You can use paper clips, blocks, or hands.

Example:

A book is 6 paper clips long.

A toy car is 3 paper clips long.

The book is longer than the toy car.

Important reminder: When measuring with objects, use the same size object each time. Do not mix big paper clips and small paper clips.

### Practice Time: Length

Use blocks, paper clips, or small toys to measure.

1. How many blocks long is your pencil?
2. How many paper clips long is your book?

3. Which is longer: your shoe or your hand?
4. Which is shorter: a crayon or a spoon?

### Time with Clocks

Important Math Word: Time helps us know when things happen.

A clock helps us measure time.

A clock can tell us:

- When to wake up
- When school starts
- When it is lunch time
- When it is bedtime

Many clocks have two hands.

The short hand points to the hour.

The long hand helps show the minutes.

For now, let us practice simple times when the long hand points straight up to 12.

If the short hand points to 3 and the long hand points to 12, the time is 3 o'clock.

If the short hand points to 7 and the long hand points to 12, the time is 7 o'clock.

### Practice Time: Time

Say the time.

1. Short hand at 2, long hand at 12.
2. Short hand at 5, long hand at 12.
3. Short hand at 9, long hand at 12.

### Check Your Answers

1. 2 o'clock
2. 5 o'clock
3. 9 o'clock

### Comparing Size

Measurement also helps us compare.

Important Math Words:

- Longer means more length.
- Shorter means less length.
- Taller means more height.
- Smaller means less size.
- Bigger means more size.

Example 6: Towers of Blocks

Tower A has 4 blocks.

Tower B has 7 blocks.

Tower B is taller.

Tower A is shorter.

Practice Time: Compare

1. A pencil is 8 blocks long. A crayon is 5 blocks long. Which is longer?
2. A stack has 3 cups. Another stack has 6 cups. Which stack is taller?
3. A small ball and a big ball are on the floor. Which one is bigger?

Check Your Answers

1. The pencil is longer.
2. The stack with 6 cups is taller.
3. The big ball is bigger.

8. Everyday Math: A Mini Problem Set

Now let us use groups, sharing, shapes, patterns, and measurement together.

Problem 1: Cookie Groups

There are 3 plates.

Each plate has 4 cookies.

How many cookies are there in all?

Step 1: Think of 3 equal groups.

Step 2: Each group has 4.

Step 3: Add  $4 + 4 + 4$ .

Step 4: The answer is 12 cookies.

#### Problem 2: Sharing Stickers

There are 12 stickers.

Four children share them equally.

How many stickers does each child get?

Step 1: Draw 4 children or 4 circles.

Step 2: Share the stickers one at a time.

Step 3: Each child gets 3 stickers.

Step 4: The answer is 3 stickers each.

#### Problem 3: Shape and Pattern

A pattern is circle, square, circle, square, circle.

What comes next?

Step 1: Look for the repeating part.

Step 2: The pattern repeats circle, square.

Step 3: After circle comes square.

Step 4: The answer is square.

#### Problem 4: Measuring at Home

Liam measures his toy train with blocks.

The train is 9 blocks long.

His toy car is 5 blocks long.

Which toy is longer?

Step 1: Compare 9 and 5.

Step 2: 9 is more than 5.

Step 3: The train is longer.

#### Your Turn: Mixed Practice

1. There are 2 boxes. Each box has 6 crayons. How many crayons are there in all?
2. Share 8 grapes equally between 2 children. How many grapes does each child get?
3. Name a shape with 3 sides.
4. What comes next: red, green, red, green, red, \_\_\_?
5. A pencil is 6 paper clips long. A marker is 8 paper clips long. Which is longer?

#### Check Your Answers

1. 12 crayons
2. 4 grapes each
3. Triangle
4. Green
5. The marker is longer.

### Friendly Recap

Great work! You learned many new math ideas in this chapter.

Here are the main ideas to remember:

- Multiplication means making equal groups.
- Repeated addition means adding the same number again and again.
- 3 groups of 2 is the same as  $2 + 2 + 2$ .
- Division means sharing equally.
- A fair share means everyone gets the same amount.
- Circles, squares, triangles, and rectangles are basic shapes.
- Sorting means putting things into groups that are alike.
- A pattern repeats in a way we can notice.
- Measurement helps us describe length, time, height, and size.
- Rulers measure length.
- Clocks measure time.
- Everyday objects like blocks and paper clips can help us measure too.

Math is all around you. You can find it in snacks, toys, clocks, shapes, songs, and games. Keep looking for equal groups, fair shares, shapes, patterns, and measurements in your day. The more you notice math, the more confident you will become.

## Multiplication as Equal Groups!



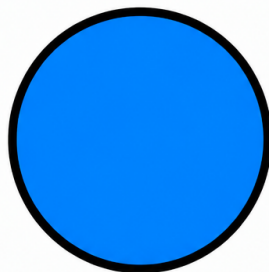
$$3 \text{ groups of } 4 = 12$$



Three plates each hold four cookies, showing three equal groups for a total of twelve cookies.



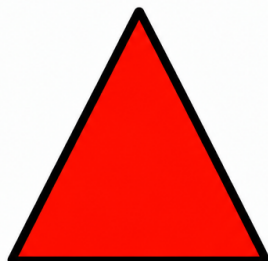
Twelve crayons are shared equally among three children, with four crayons for each child.



**Circle**



**Square**



**Triangle**



**Rectangle**

Four labeled shapes are shown: a circle, a square, a triangle, and a rectangle.

length

time

height

*A pencil is measured with a ruler, a clock shows time, and two objects are compared by height.*

# Chapter 5

## Using Math Every Day: Beginner Word Problems and Final Review

### Chapter 5: Using Math Every Day: Beginner Word Problems and Final Review

In this final chapter, you will discover how math helps you every day. Math is not only something you do on paper. You use math when you count snacks, compare toys, share treats, find shapes, follow patterns, and measure things.

You have learned many important beginner math ideas. Now we will put them together to solve simple problems with confidence.

#### Math Is All Around You

Math is the study of numbers, shapes, patterns, and measuring. It helps us understand how many, how much, how long, how big, and what comes next.

You can find math in many places:

- At home: counting plates, sharing cookies, measuring water, looking at the clock
- At school: counting pencils, comparing books, making patterns, lining up
- At play: counting blocks, sharing toys, sorting shapes, keeping score
- At a store: counting coins, comparing prices, adding items

Important math word: Count

Count means to say numbers in order to find how many there are.

Example:

Mia has 1, 2, 3, 4 crayons.

Mia has 4 crayons.

Practice:

1. Count your fingers on one hand. How many are there?
2. Count 6 steps as you walk across the room.
3. Count 4 things near you.

#### Everyday Counting Problems

Counting helps us answer the question, How many?

Worked Example 1:

There are apples on a plate. You count them: 1, 2, 3, 4, 5.

How many apples are there?

Answer: There are 5 apples.

Worked Example 2:

Sam puts toy cars in a row. He counts: 1, 2, 3, 4, 5, 6, 7.

How many toy cars does Sam have?

Answer: Sam has 7 toy cars.

Try It:

1. Count 3 spoons. How many spoons did you count?
2. Count 8 blocks. How many blocks did you count?
3. Count 10 claps. What number did you stop on?

Everyday Comparing Problems

Sometimes math helps us compare numbers.

Important math word: Compare

Compare means to look at two numbers or groups and decide which is more, which is less, or if they are equal.

Important math word: Equal

Equal means the same amount.

Worked Example:

Lena has 4 stickers. Ben has 6 stickers.

Who has more stickers?

Think:

6 is more than 4.

Answer: Ben has more stickers.

Worked Example:

A red basket has 5 balls. A blue basket has 5 balls.

Which basket has more?

Think:

5 and 5 are equal.

Answer: Neither basket has more. They are equal.

Try It:

1. Tom has 3 pencils. Ana has 5 pencils. Who has more?
2. One shelf has 7 books. Another shelf has 4 books. Which shelf has fewer?
3. A cup has 6 beads. A box has 6 beads. Are they equal?

### A Simple Way to Solve Word Problems

A word problem is a math question told with words. Word problems may talk about snacks, toys, animals, books, or other everyday things.

Do not worry if a word problem looks long. Use these steps.

#### Five Steps for Beginner Word Problems:

1. Read the problem.
2. Find the numbers.
3. Choose what to do.
4. Solve.
5. Check your answer.

Let us learn each step.

Step 1: Read the problem.

Read slowly. Ask, What is happening?

Step 2: Find the numbers.

Look for numbers in the problem.

Step 3: Choose what to do.

Ask yourself:

- Do I need to count?
- Do I need to compare?
- Do I need to add?
- Do I need to subtract?
- Do I need to make equal groups?
- Do I need to share equally?

Step 4: Solve.

Use fingers, drawings, objects, or numbers.

Step 5: Check your answer.

Ask, Does my answer make sense?

### Addition Word Problems

Important math word: Add

Add means to put numbers or groups together to find how many in all.

When a problem asks how many in all, you may need to add.

Worked Example:

Nora has 3 pencils. Her teacher gives her 2 more pencils. How many pencils does Nora have now?

1. Read: Nora has pencils, then gets more.
2. Find the numbers: 3 and 2.
3. Choose: More pencils means add.
4. Solve:  $3 + 2 = 5$ .
5. Check: 5 is more than 3, so it makes sense.

Answer: Nora has 5 pencils.

You can also draw it:

3 pencils: |||

2 more pencils: ||

All together: |||||

That makes 5 pencils.

Try It:

1. Eli has 4 blocks. He gets 1 more block. How many blocks does he have now?
2. A plate has 2 cookies. Dad adds 3 cookies. How many cookies are on the plate?
3. There are 5 birds in a tree. 2 more birds fly in. How many birds are in the tree now?

### Subtraction Word Problems

Important math word: Subtract

Subtract means to take away or find how many are left.

When a problem says something goes away, is eaten, is used, or is taken, you may need to subtract.

Worked Example:

Jay has 6 crackers. He eats 2 crackers. How many crackers are left?

1. Read: Jay has crackers and eats some.
2. Find the numbers: 6 and 2.
3. Choose: Eating crackers takes them away, so subtract.
4. Solve:  $6 - 2 = 4$ .
5. Check: 4 is less than 6, so it makes sense.

Answer: Jay has 4 crackers left.

Try It:

1. Kim has 5 balloons. 1 balloon pops. How many balloons are left?
2. There are 8 apples. 3 apples are eaten. How many apples are left?
3. Omar has 7 toy cars. He gives 2 away. How many toy cars does he have now?

Equal Groups: Multiplication as Repeated Addition

Important math word: Equal groups

Equal groups are groups with the same number in each group.

Multiplication begins with equal groups. For now, think of multiplication as adding the same number again and again.

Worked Example:

There are 3 plates. Each plate has 2 cookies. How many cookies are there in all?

1. Read: There are equal groups of cookies.
2. Find the numbers: 3 plates and 2 cookies on each plate.
3. Choose: Equal groups means repeated addition.
4. Solve:  $2 + 2 + 2 = 6$ .
5. Check: Three groups of 2 makes 6.

Answer: There are 6 cookies in all.

Try It:

1. There are 2 bags. Each bag has 3 marbles. How many marbles are there in all?
2. There are 4 cups. Each cup has 2 straws. How many straws are there in all?
3. There are 3 boxes. Each box has 1 toy. How many toys are there in all?

Sharing Equally: Division Begins Here

Important math word: Share equally

Share equally means each person or group gets the same amount.

Division begins with sharing. You do not need big division facts yet. You can draw pictures or use objects.

Worked Example:

There are 8 grapes. Two children share them equally. How many grapes does each child get?

1. Read: Grapes are shared equally.
2. Find the numbers: 8 grapes and 2 children.
3. Choose: Sharing equally.
4. Solve: Give 1 grape to each child again and again until all are gone.

Child 1: 4 grapes

Child 2: 4 grapes

5. Check: 4 and 4 make 8.

Answer: Each child gets 4 grapes.

Try It:

1. Share 6 crackers equally between 2 friends. How many does each friend get?
2. Share 4 stickers equally between 2 children. How many does each child get?
3. Share 9 blocks equally into 3 boxes. How many blocks go in each box?

Place Value Review: Tens and Ones

Important math word: Place value

Place value tells what a digit is worth because of where it is in a number.

In beginner math, we learn about tens and ones.

Example:

The number 14 has:

- 1 ten
- 4 ones

14 means 10 and 4 more.

Worked Example:

What does 23 mean?

23 has:

- 2 tens
- 3 ones

2 tens are 20. Add 3 ones. That makes 23.



Try It:

1. In 12, how many tens? How many ones?
2. In 35, how many tens? How many ones?
3. What number has 4 tens and 2 ones?

## Shape Review

Important math word: Shape

A shape is the form or outline of something.

You can find shapes everywhere.

- Circle: round like a ball or plate
- Square: 4 equal sides, like some tiles
- Rectangle: 4 sides, like a door or book
- Triangle: 3 sides, like a slice of pizza

Worked Example:

A window has 4 sides. It is longer than it is tall. What shape might it be?

Answer: It might be a rectangle.

Try It:

1. What shape is a wheel?
2. What shape has 3 sides?
3. Find one rectangle near you.

## Pattern Review

Important math word: Pattern

A pattern is something that repeats in a regular way.

Patterns help us guess what comes next.

Worked Example:

Red, blue, red, blue, red, \_\_\_\_

What comes next?

The pattern repeats red, blue.

Answer: Blue comes next.

Worked Example:

2, 4, 6, 8, \_\_\_\_

The numbers count by 2s.

Answer: 10 comes next.

Try It:

1. Clap, tap, clap, tap, clap, \_\_\_\_ . What comes next?
2. Circle, square, circle, square, \_\_\_\_ . What comes next?
3. 5, 6, 7, 8, \_\_\_\_ . What comes next?

## Measurement Review

Important math word: Measure

Measure means to find the size, length, height, weight, or amount of something.

You can measure with rulers, cups, hands, steps, or blocks.

Worked Example:

A pencil is 6 blocks long. An eraser is 2 blocks long. Which is longer?

Think:

6 is more than 2.

Answer: The pencil is longer.

Worked Example:

A bottle holds 3 cups of water. A bowl holds 5 cups of water. Which holds more?

Think:

5 is more than 3.

Answer: The bowl holds more.

Try It:

1. Which is longer, 4 steps or 7 steps?
2. Which is shorter, 2 blocks or 6 blocks?
3. Find something taller than your hand.

## Mixed Word Problems

Now let us mix different kinds of math. Remember the five steps:

1. Read.
2. Find the numbers.

3. Choose what to do.
4. Solve.
5. Check.

Problem 1:

Sara has 6 shells. Leo has 4 shells. Who has more shells?

Numbers: 6 and 4.

Choose: Compare.

Solve: 6 is more than 4.

Answer: Sara has more shells.

Problem 2:

There are 3 cats on a rug. 2 more cats come. How many cats are on the rug now?

Numbers: 3 and 2.

Choose: Add.

Solve:  $3 + 2 = 5$ .

Answer: There are 5 cats.

Problem 3:

A box has 9 crayons. Tara takes out 3 crayons. How many crayons are left in the box?

Numbers: 9 and 3.

Choose: Subtract.

Solve:  $9 - 3 = 6$ .

Answer: 6 crayons are left.

Problem 4:

There are 2 baskets. Each basket has 5 oranges. How many oranges are there in all?

Numbers: 2 and 5.

Choose: Equal groups.

Solve:  $5 + 5 = 10$ .

Answer: There are 10 oranges.

Problem 5:

There are 6 stickers. Three children share them equally. How many stickers does each child get?

Numbers: 6 and 3.

Choose: Share equally.

Solve: Give 1 sticker to each child, then another 1 sticker to each child. Each child gets 2.

Answer: Each child gets 2 stickers.

## Final Review: Your Math Toolbox

You now have a beginner math toolbox. These are the math tools you can use:

- Numbers help us name how many.
- Counting helps us find a total.
- Comparing helps us find more, less, or equal.
- Place value helps us understand tens and ones.
- Addition helps us put groups together.
- Subtraction helps us take away.
- Equal groups help us begin multiplication.
- Sharing equally helps us begin division.
- Shapes help us describe objects.
- Patterns help us know what comes next.
- Measurement helps us find length, height, weight, or amount.

## Final Practice Activities

### Activity 1: Count and Compare

A jar has 8 buttons. A cup has 5 buttons.

1. Which has more buttons?
2. Which has fewer buttons?

### Activity 2: Add

Max has 4 books. He gets 3 more books.

How many books does Max have now?

### Activity 3: Subtract

There are 10 grapes. You eat 4 grapes.

How many grapes are left?

### Activity 4: Tens and Ones

Write the tens and ones.

1.  $16 = \underline{\quad}$  ten and  $\underline{\quad}$  ones
2.  $28 = \underline{\quad}$  tens and  $\underline{\quad}$  ones
3.  $31 = \underline{\quad}$  tens and  $\underline{\quad}$  one

### Activity 5: Equal Groups

There are 3 cups. Each cup has 2 pencils.

How many pencils are there in all?

### Activity 6: Share Equally

Share 8 blocks equally between 4 children.  
How many blocks does each child get?

#### Activity 7: Shapes

Name the shape.

1. A clock face is shaped like a \_\_\_\_
2. A slice of pizza may look like a \_\_\_\_
3. A book may look like a \_\_\_\_

#### Activity 8: Patterns

What comes next?

1. Star, moon, star, moon, \_\_\_\_
2. 1, 2, 3, 4, \_\_\_\_
3. Green, green, yellow, green, green, yellow, \_\_\_\_

#### Activity 9: Measurement

Use your hands or steps.

1. Measure a table with your hands. About how many hands long is it?
2. Measure a room with your steps. About how many steps across is it?
3. Find something shorter than your foot.

#### Answer Check

##### Activity 1:

1. The jar has more buttons.
2. The cup has fewer buttons.

##### Activity 2:

$4 + 3 = 7$ . Max has 7 books.

##### Activity 3:

$10 - 4 = 6$ . There are 6 grapes left.

##### Activity 4:

1.  $16 = 1$  ten and 6 ones
2.  $28 = 2$  tens and 8 ones
3.  $31 = 3$  tens and 1 one

##### Activity 5:

$2 + 2 + 2 = 6$ . There are 6 pencils.

##### Activity 6:

Each child gets 2 blocks.

Activity 7:

1. Circle
2. Triangle
3. Rectangle

Activity 8:

1. Star
2. 5
3. Green

Activity 9:

Your answers may be different because rooms and objects can be different sizes.

Friendly Recap

Great work! You have finished the final chapter of Math Newbie.

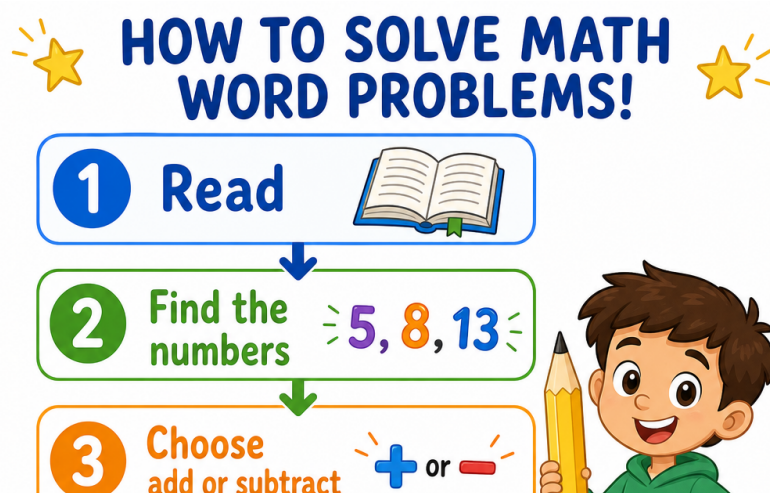
You learned that math is part of everyday life. You can use math to count, compare, add, subtract, make equal groups, share equally, find shapes, notice patterns, and measure objects.

You also learned a simple way to solve word problems:

1. Read the problem.
2. Find the numbers.
3. Choose what to do.
4. Solve.
5. Check your answer.

Remember, math gets easier with practice. You do not have to be fast right away. Go slowly, use pictures or objects, and think carefully. Every time you solve a problem, your math brain grows stronger.

You are no longer starting from no knowledge at all. You are now a confident beginner. Keep counting, keep noticing, and keep trying. Math is everywhere, and you are ready to use it!



*A five-step flowchart shows how to solve a word problem: read, find the numbers, choose an action, solve, and check.*